Reviewer report: scikit-learn 1.5.2 to 1.6.0rc1 (contains PR #29677)

Run `golden-sklearn160rc1-imblearn0124`, tested against imbalanced-learn 0.12.4. Report generated 2026-10-10 05:53 IST by reports/generate.py 1.0 (from Jonathan's render_report.py 0.1).

NEEDS ATTENTION

Recommended action: **Reject** Confidence: **High**

All required checks and evals passed, 3 runs per side gave identical results (stability old 1.0, new 1.0), 3 of 3 golden cases hold, tests ran without network.

Scope: this run compares two released versions, not a single PR's commits. Any change between the two versions can cause what is shown here.

Summary in plain English

- **What it says it does:** PR #29677 "API Revamp estimator tags" replaces the old estimator tags with a new tags API (PR title, github.com/scikit-learn/scikit-learn/pull/29677, verified via the GitHub API on 2026-10-10).
- **What we observed:** imbalanced-learn 0.12.4 can no longer be imported with scikit-learn 1.6.0rc1: 46 test module(s) fail to load and 2243 test(s) that passed before cannot run. Same result in all 3 runs on each side.
- **What you should do:** Do not approve as is. The change breaks imbalanced-learn; see 'What changed in behavior' for what fails.

Claims vs reality

#	Claim (source)	Status	What we observed	Log line
C1	Revamps the estimator tags API (PR #29677 title)	not testable	the dependent's tests show whether existing code still works; they cannot show the new tags API is correct	pytest-new-1.log:28 «_safe_tags»
-	Change not in the PR description (this run)	unmentioned change	46 test module(s) cannot be imported: ImportError: cannot import name '_safe_tags' from 'sklearn.utils._tags' (linked to C1 by the PR #29677 diff, which deletes _safe_tags (checked in git on 2026-10-10))	pytest-new-1.log:28 «ImportError: cannot import name '_safe_tags' from 'sklearn.utils._tags»

Citation check (PASS): 2 claim(s), every one backed by a real log line

What changed in behavior

Import failure: 46 test module(s) cannot load

Item	Value
Error	ImportError: cannot import name '_safe_tags' from 'sklearn.utils._tags'
Log line	pytest-new-1.log:28 «ImportError: cannot import name '_safe_tags' from 'sklearn.utils._tags»
Before (old)	package import: imblearn: ok
After (new)	imblearn: ImportError: cannot import name '_safe_tags' from 'sklearn.utils._tags' (/workspace/blast-radius-research/research/runs/golden-sklearn160rc1-imblearn0124/venv-new/lib/python3.13/site-packages/sklearn/utils/_tags.py)
Tests that passed before and can no longer run	2243
Affected modules (first 8)	combine.tests.test_smote_enn, combine.tests.test_smote_tomek, datasets.tests.test_imbalance, datasets.tests.test_zenodo, ensemble.tests.test_bagging, ensemble.tests.test_easy_ensemble, ensemble.tests.test_forest, ensemble.tests.test_weight_boosting, and 38 more
Full lists	class-import_failure.txt, class-lost_tests.txt

What went right

Status	Check	Detail
pass	Correct versions installed	old scikit-learn==1.5.2, new scikit-learn==1.6.0rc1, imbalanced-learn==0.12.4 on both sides (freeze + import probe)
pass	Only the upstream version differs between sides	freeze diff: ['scikit-learn==1.5.2', 'scikit-learn==1.6.0rc1']
pass	Isolated environments	both venvs created by uv without system site-packages (checked sys.prefix and sys.path)
pass	Thread count pinned to 1	OMP_NUM_THREADS, OPENBLAS_NUM_THREADS, MKL_NUM_THREADS, NUMEXPR_NUM_THREADS, VECLIB_MAXIMUM_THREADS, BLIS_NUM_THREADS = 1 in both envs
pass	All runs completed	3 runs per side requested (minimum 3); JUnit XML present and no suite timeout on every run
pass	Old side healthy (baseline is trustworthy)	dependent imports on old: {'imblearn': 'ok'}; old side passed [2243, 2243, 2243] tests with collection errors [0, 0, 0]
pass	Repeatable results	share of tests with the same outcome in every run: old 1.0, new 1.0 (advisory threshold 0.99)
pass	No secrets in logs	scrubbed env; no credential-like strings found in logs
pass	Record format valid	record matches harness/schema/observation.schema.json
pass	Runs per side	3 (minimum 3), alternating old and new
pass	Guardrails respected	tests ran with no network; nothing was posted or filed by the runner

Status	Check	Detail
pass	Matches the prediction written before the run	break: imbalanced-learn 0.12.4 cannot be imported on 1.6.0rc1 (ImportError on sklearn.utils._tags._safe_tags). Expect old 2238 passed (pytest summary), new 0 passed / 525 errors, import_failure 46 modules, lost_tests 2243, real_break 0.
pass	Nathan's evals	7 of 8 eval checks passed (see 'Evals and guardrails')
pass	Every claim cites a log line	2 claim(s), every one backed by a real log line

What went wrong and how to fix it

Issues in the PR or release

Issue	How to fix
imbalanced-learn cannot be imported on the new side (ImportError: cannot import name '_safe_tags' from 'sklearn.utils._tags')	The change removed or renamed something imbalanced-learn uses. Either keep a deprecated alias in scikit-learn for one release, or imbalanced-learn must release a fix before this ships.

Issues in our test environment

None found.

Trust checks

Check	Status	Detail
Required checks (gates)	7 of 7 passed	all passed
Advisory checks	2 of 2 passed	all passed
Evals (Nathan)	APPROVE	no failures or warnings
Golden cases (stored records)	3 of 3 hold	checked against the stored golden records (harness 1.0) with the eval gate's code, not re-run for this report. This run is itself one of the golden cases, so this check is partly circular for it
Golden cases (re-run)	not available	no golden re-run recorded in this record (golden_check)
Report citations	pass	2 claim(s), every one backed by a real log line
Flaky rate	0 of 2295 (0.0%)	stability old 1.0, new 1.0
Network during tests	off	install: allowed (uv from PyPI/nightly index, git clone of the tag); tests: no network (unshare -rn, private network namespace)

Check	Status	Detail
Threads	pinned to 1	OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1, MKL_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1, VECLIB_MAXIMUM_THREADS=1, BLIS_NUM_THREADS=1
Machine	box	Intel(R) Xeon(R) Processor, 8 CPUs, 15.6 GB RAM, Linux x86_64, CI: not available
Report matches the record	consistent	break and import-failure counts recomputed from class-*.txt and JUnit equal the record
Prediction before the run	yes	break: imbalanced-learn 0.12.4 cannot be imported on 1.6.0rc1 (ImportError on sklearn.utils._tags._safe_tags). Expect old 2238 passed (pytest summary), new 0 passed / 525 errors, import_failure 46 modules, lost_tests 2243, real_break 0.

Golden case	Status	Detail
golden-sklearn160rc1-imblearn-import-failure	holds	golden-sklearn160rc1-imblearn-import-failure: all expected values hold
golden-pandas300rc0-upsetplot-cow	holds	golden-pandas300rc0-upsetplot-cow: all expected values hold
golden-clean-sklearn151-152-imblearn	holds	golden-clean-sklearn151-152-imblearn: all expected values hold

Evals and guardrails (Nathan)

Independent checks of this record by evals/checks.py and guardrails/schema.py against the shared schema `blast-radius/observation/v1` (written against rev 1 (sha256 5a3c3d36b773...); valid under current rev 2 (additive)). Eval recommendation: **APPROVE**. Any FAIL here makes the verdict COULD NOT VERIFY.

Check	Status	Detail	How to fix
schema	pass	record matches the shared schema blast-radius/observation/v1, mode v1 (schema reference verified by sha256)	-
versions_installed	pass	scikit-learn 1.5.2 -> 1.6.0rc1; env differs only in scikit-learn	-
runs_per_side	pass	3 alternating runs per side, identical counts within each side	-
single_thread	pass	OMP/OPENBLAS/MKL_NUM_THREADS=1	-
matches_expected	pass	outcome matches the prediction written before the run (break)	-
golden:golden-sklearn160rc1-imblearn-import-failure	pass	golden-sklearn160rc1-imblearn-import-failure: all expected values hold	-
golden:golden-sklearn160rc1-imblearn-import-failure:import_message	pass	import error mentioning '_safe_tags' found in new-side import probe (imblearn: ImportError: cannot import name '_safe_tags' from 'sklearn.utils._tags' (/worksp); break signature (ImportError: cannot import name '_safe_tags' from 'sklearn.utils._tags' (<path>)	-
llm_citations	skip	no Nemotron explanations in this record yet	-

Check	Status	Detail	How to fix
report_citations	pass	2 claim(s), every one backed by a real log line	-

Reviewer checklist

☐ I read the verdict, the recommended action and the confidence.
☐ I checked the claims vs reality table for any 'false' or 'unmentioned change' rows, and opened at least one cited log line.
☐ I looked at the largest behavior-change group and its example error.
☐ I confirmed the required checks and Nathan's evals passed (Trust checks, Evals and guardrails).
☐ I am satisfied the environment issues (if any) do not change the decision.

Decision: ☐ Approve ☐ Approve with notes ☐ Reject ☐ Re-run Reviewer: _____ Date: _____

Rerun command

From the blast-radius repo root. The wrapper forces 3 alternating runs per side and the single-thread env, records machine_type, writes under ./data and never posts anything:

```
BR_RUNNER=/workspace/blast-radius-research/harness/br_run.py python -m harness.run \
  --upstream scikit-learn \
  --old 1.5.2 \
  --new 1.6.0rc1 \
  --dependent imbalanced-learn \
  --dep-version 0.12.4 \
  --tests wheel:imblearn \
  --extra-pkgs 'numpy==2.1.3 scipy==1.14.1 pandas==2.2.3' \
  --cpus 0-3 \
  --expected 'break: imbalanced-learn 0.12.4 cannot be imported on 1.6.0rc1 (ImportError on sklear
n.utils._tags._safe_tags). Expect old 2238 passed (pytest summary), new 0 passed / 525 errors, imp
ort_failure 46 modules, lost_tests 2243, real_break 0.' \
  --expected-class break \
  --id golden-sklearn160rc1-imblearn0124-rerun
```

Needs Linux x86_64, uv 0.12 or newer, git and unshare.

Evidence links

What	Link (relative to this report)
Where the description comes from	PR title, github.com/scikit-learn/scikit-learn/pull/29677, verified via the GitHub API on 2026-10-10
Claim C1	pytest-new-1.log:28 «_safe_tags»
Unmentioned change	pytest-new-1.log:28 «ImportError: cannot import name '_safe_tags' from 'sklearn.utils._tags»

What	Link (relative to this report)
pytest summary, old side run 1	pytest-old-1.log:123 «2238 passed, 33 skipped, 1 xfailed, 5 xpassed, 62 warnings in 42.45s»
pytest summary, new side run 1	pytest-new-1.log:88681 «525 errors in 14.44s»
Full record	record.json
Run arguments	args.json
Setup log	setup.log
Installed packages, old	freeze-old.txt
Installed packages, new	freeze-new.txt
Package difference	freeze-diff.txt
pytest logs	pytest-new-1.log, pytest-new-2.log, pytest-new-3.log, pytest-old-1.log, pytest-old-2.log, pytest-old-3.log
JUnit XML	junit-new-1.xml, junit-new-2.xml, junit-new-3.xml, junit-old-1.xml, junit-old-2.xml, junit-old-3.xml
Test lists per class	class-already_broken.txt, class-fixed_by_new.txt, class-flaky.txt, class-import_failure.txt, class-lost_tests.txt, class-real_break.txt

Appendix

Commits and versions

Item	Value
PR	not available
PR base / head / merge commit	not available
Upstream	scikit-learn: old 1.5.2, new 1.6.0rc1 (requested 1.6.0rc1)
Upstream commits (old / new)	PyPI release wheels (no commit recorded)
Dependent	imbalanced-learn 0.12.4
Tests from	wheel (tests shipped in the installed package: imblearn); selection: imblearn
Python	3.13.5
Environment difference	scikit-learn==1.5.2, scikit-learn==1.6.0rc1
Runner	harness 1.0, 4 workers, import mode prepend, per-test timeout 300 s, memory cap 6.0 GB
Record schema	v1; written against rev 1 (sha256 5a3c3d36b773...); valid under current rev 2 (additive)
Run by	Jonathan on box, 2026-10-10 05:36 IST
Hand check	not available
Upstream issue	not available
Per-test outcomes	not available

Item	Value
AI explanation	not yet wired

Timings

Total wall time 189.1 s; mean test time old 44.8 s, new 15.5 s.

Side	Run	Exit	Seconds	pytest summary
old	1	0	43.22	2238 passed, 33 skipped, 1 xfailed, 5 xpassed, 62 warnings in 42.45s
new	1	1	14.98	525 errors in 14.44s
old	2	0	47.55	2238 passed, 33 skipped, 1 xfailed, 5 xpassed, 62 warnings in 46.64s
new	2	1	16.09	525 errors in 15.52s
old	3	0	43.68	2238 passed, 33 skipped, 1 xfailed, 5 xpassed, 62 warnings in 42.85s
new	3	1	15.43	525 errors in 14.87s

Glossary

Term	Meaning
Old side / new side	The same dependent library tested twice: once with the old version of the upstream library, once with the new one. Nothing else differs.
Upstream	The library being changed (for example pandas). The PR or release under review belongs to it.
Dependent	A library that uses the upstream. Its own tests are our test subject.
Real break	A test that passed in every old run and failed in every new run. This is a behavior change caused by the upstream change.
Import failure	A test module that cannot even be loaded on the new side, so none of its tests can run. Counted per module.
Lost tests	Tests that passed before but could not run after, because their module failed to load. Not counted as real breaks.
Already broken	Tests that failed before the change too. Not caused by the change.
Flaky	Tests whose result changes between runs on the same side. They are not trusted either way.
Error signature	The error message with numbers and file paths removed, so identical failures group together.
Gate	An automatic check by the runner that the run itself is trustworthy (right versions, same environment, enough runs). Required gates must pass for the result to count.
Eval	Nathan's independent check of the record (schema, versions, runs per side, threads, golden cases, citations). Any eval failure means the run cannot be trusted.
Golden case	A known example with a known answer. If the tool still gets the golden cases right, its answers on new cases are more believable.

Term	Meaning
Stability	Share of tests that gave the same result in every run on one side. 1.0 means fully repeatable.
Citation	A file name, line number and quoted text from a run log that backs a statement in the report.
JUnit XML	The machine-readable test result file pytest writes for each run.
Freeze	The exact list of installed packages and versions on each side.